Name	CTG

YISHUN JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2018

BIOLOGY **9744/04**

HIGHER 2 **16 AUG 2018**
Paper 4

2 hours 30 minutes

Candidates answer on the Question Paper

Additional material: As listed in the Confidential Instructions

READ THESE INSTRUCTIONS FIRST

Write your name and CTG in the spaces at the top of this page and on all the work you hand in.
Give details of the practical shift and the laboratory, where appropriate, in the boxes provided.
Write in dark blue or black pen only.

You may use a HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	/15
2	/30
3	/10
Total	**/55**

This paper consists of **20** printed pages.

Before you proceed, read carefully through the **whole** of the question paper.

Plan the use of the **two and a half hours** to make sure that you finish all the work that you would like to do.

You will **gain marks** for recording your results according to the instructions.

1 You are required to estimate the water potential of potato tissue.

When a piece of potato is put into a sucrose solution, as shown in Fig. 1.1, water will move by osmosis into and out of the potato cells. The net direction of movement of water depends on the difference in water potential between the potato cells and the sucrose solution.

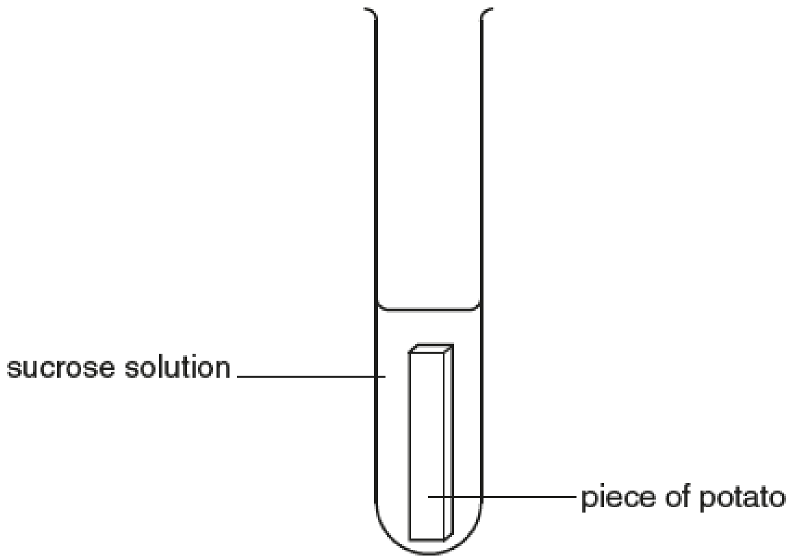

Fig. 1.1

(a) (i) State a hypothesis that can be used to estimate the water potential of the potato tissue.

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……………………..……………………………………………………………………[1]

A piece of potato is left in a sucrose solution for 15 minutes, to allow time for osmosis to take place.

The concentration of the sucrose solution after 15 minutes may be different from the original concentration.

After 15 minutes a blue dye is added to the sucrose solution to make a blue solution. The blue dye does not affect the concentration of the sucrose solution.

A small volume of this blue solution is then released into a known concentration of sucrose solution as shown in Fig. 1.2.

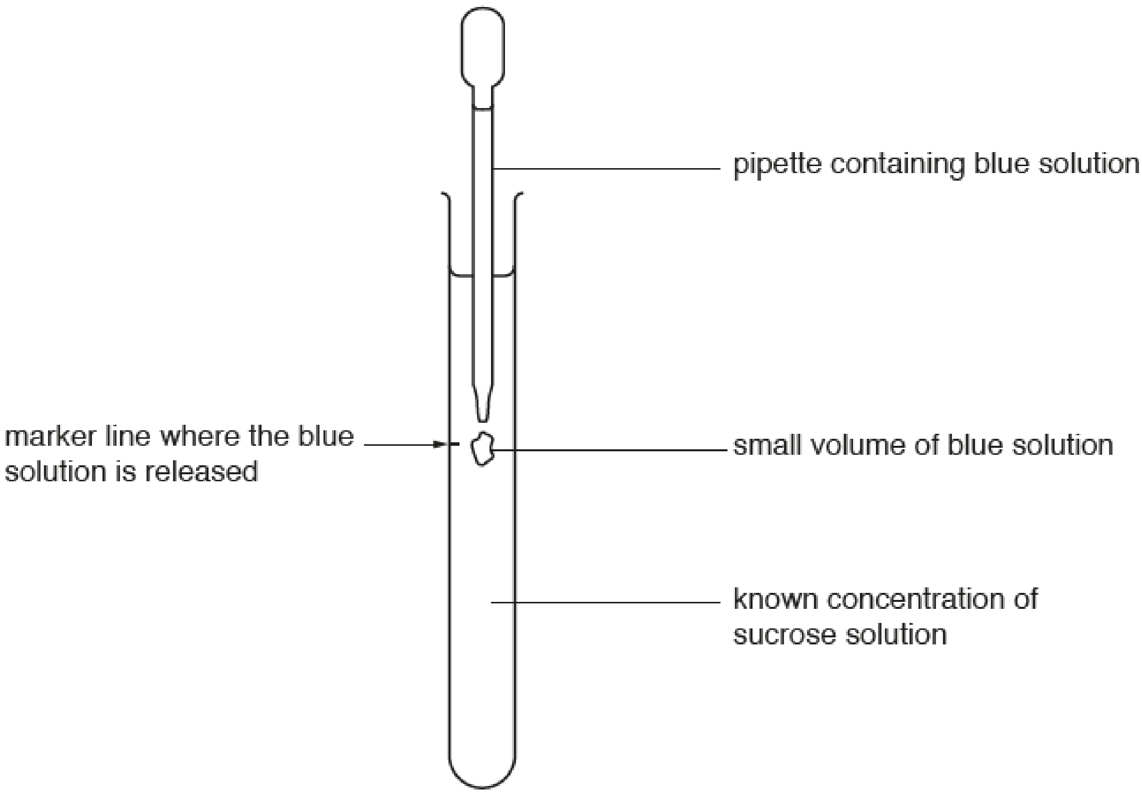

Fig. 1.2

The pipette is removed immediately after the blue solution is released.

(ii) Describe how you can use different concentrations of sucrose solutions to predict the behaviour of the blue solution.

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(b) You are required to investigate osmosis in potato tissue so that you can estimate the water potential of potato cells.

You are provided with:

labelled	contents	hazard	Volume / cm^3
S1	1.00 mol dm^{-3} sucrose solution	None	70
S2	0.50 mol dm^{-3} sucrose solution	None	70
S3	0.25 mol dm^{-3} sucrose solution	None	70

labelled	contents
P	5 potato cylinders, each measuring 4 cm in length

Read step 1 to step 11 before proceeding.

1. Measure 4.5 cm from the bottom of each **boiling tube** and put a mark, as shown in Fig. 1.3.

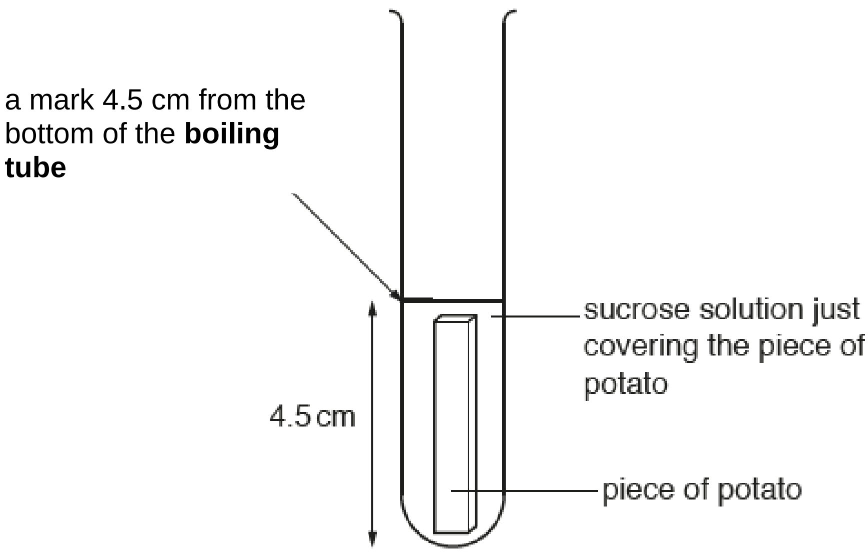

Fig. 1.3

(i) Decide on the length of each piece of potato cylinder you will use, so that the volume of sucrose solution just covers the piece of potato, as shown in Fig. 1.3.

Length………………………………………..[1]

2. Cut enough pieces of potato to put into the three sucrose concentrations that you were provided.

3. Put the pieces of potato on a paper towel to remove any excess fluid.

4. Put one piece of potato into each of the boiling tubes from step 1.

5. Put 1.00 mol dm^{-3} sucrose solution, **S1**, into one of the boiling tubes up to the mark made in step 1.

6. Repeat step 5 with the remaining concentrations of sucrose solutions.

7. Start timing and leave for 15 minutes.

While you are waiting for 15 minutes, continue with step 8 to step 10 and continue with the other questions.

8. Measure 6 cm from the **top** of each of the **test tubes** and put a mark, as shown in Fig. 1.4.

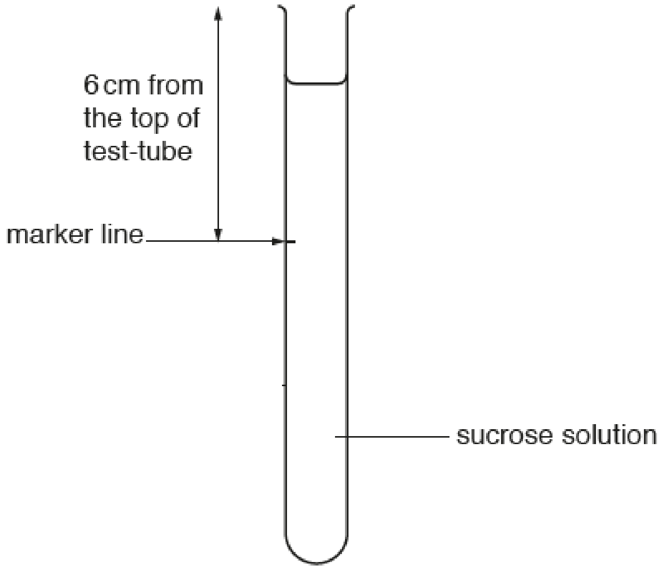

Fig. 1.4

9. Put 15 cm^3 of 1.00 mol dm^{-3} sucrose solutions, **S1**, into one of the **test tubes** from step 8.

10. Repeat step 9 with the remaining concentrations of sucrose solution provided.

You are provided with:

labelled	contents	hazard	Volume / cm^3
M	methylene blue solution	none	15

11. After leaving the pieces of potato for 15 minutes, remove the potato cylinder from the boiling tube containing 1.00 mol dm^{-3} sucrose solution and set aside.

12. Put a drop of **M** into the boiling tube containing 1.00 mol dm^{-3} sucrose solution, **S1**. Gently shake the boiling tube to mix **M** with the sucrose solution.

13. Repeat step 11 and 12 for the remaining concentrations of sucrose solution.

(ii) Describe how you will determine the rate of osmosis in the potato cylinders and state the appropriate unit of measurement for the method you described.

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Rate of osmosis ……………………………………..….[3]

(iii) Calculate the rate of osmosis in the potato cylinders set aside in step 11 using the method you describe in **(b)(ii)** and record your results in an appropriate format in the space below.

[3]

Read step 14 to step 16 before proceeding.

14. Use a pipette to remove a sample of the blue solution from the **boiling tube** containing 1.00 mol dm–3 sucrose solution, **S1**.

You will now use the test-tubes as in Fig. 1.4.

15. Put the end of the pipette into the **test-tube** containing 1.00 mol dm–3 concentration of sucrose solution, **S1**. This should be level with the marker line on the test-tube as shown in Fig. 1.1 on page 2.

16. Release a small volume of the blue solution, then immediately remove the pipette from the test-tube.
It is possible to repeat step 16 without having to replace this sucrose solution.

17. Repeat step 14 to step 16 for the other concentrations of sucrose solution.

(iv) Record your observations for the behaviour of the blue solution from the different test tubes in an appropriate table in the space below.

[3]

(v) Using your results in **(b)(iii)** and **(b)(iv)**, estimate the concentration of sucrose solution with a water potential equal to the water potential of the potato tissue.

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(vi) Identify **one** significant source of error in this investigation.

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[Total: 15]

2 The fruit of the cotton plant contains seeds surrounded by fibres known as lint. During processing, the lint is separated from the seeds and made into cotton fibres. The lint may stick to the processing equipment if reducing sugars or non-reducing sugars are attached to it. The quantity of sugar present determines the stickiness of the lint.

Different samples of lint were soaked for 24 hours in three beakers of water, **C1** to **C3**. The same volume of water and mass of lint was used in each beaker.

You are required to :

- make a serial dilution of 1% reducing sugar solution **R;**
- carry out tests on each concentration of reducing sugar;
- carry out tests on **C1** to **C3**;
- determine the type and concentration of sugars in **C1** to **C3**.

You are provided with:

labelled	**contents**	**hazard**	**Volume/cm^3**
R	1.0% reducing sugar solution	none	50
C1	sample	None	20
C2	sample	None	20
C3	sample	None	20
W	distilled water	None	100
Benedict's	Benedict's solution	Harmful	20

It is recommended that you wear suitable eye protection. If Benedict's solution comes into contact with your skin, wash it off immediately under cold water.

1. Set up a water-bath and heat the water to a suitable temperature to test for reducing sugars using the Benedict's test.

(a) (i) State the temperature you will need to maintain in the water-bath to carry out the Benedict's test.

Temperature ....[1]

You are required to carry out a **serial** dilution of the 1.0% reducing sugar solution, **R**, which reduces the concentration **by half** between each successive dilution. This will provide you with a set of reducing sugar solutions of known concentrations.

After the serial dilution is completed, you will need to have 10 cm^3 of each concentration available for use.

(ii) Complete Table 2.1 to show how you prepare the concentration of glucose solutions, **R2**, **R3**, **R4** and **R5,** and how you will set up a control.

Table 2.1

	R1	**R2**	**R3**	**R4**	**R5**
Concentration of reducing sugar solution / %	1.0				
Label of reducing sugar solution to be diluted		R1			
Volume of reducing sugar solution to be diluted / cm^3					
Volume of distilled water **W**, to make the dilution / cm^3		10.0	10.0	10.0	10.0
Description of the control: ……………………………………………………………………………………………………… ………………………………………………………………………………………………………					

[3]

*Read steps **2 to 4** and prepare a table in **3(a)(iv)** before proceeding.*

2. Prepare all the concentrations of reducing sugar solution shown in Table 2.1, in the beakers provided.

(iii) You will need to carry out the Benedict's test on each of the different concentrations of reducing sugar solution and on 2 cm^3 of each of **C1**, **C2** and **C3**.

You will be recording the time taken for the first appearance of a colour change.

State the volume of Benedict's solution and the volume of each of the concentrations of reducing sugar solution you will use for each test.

volume of Benedict's solution ....

volume of each concentration of reducing sugar solution...................................

[1]

3. Using the volumes you decided in **(a)(iii)**, carry out the Benedict's test on the reducing sugar solutions of different concentrations shown in Table 2.1.

   Record, in **(a)(iv)**, the time taken for the first appearance of a colour change. If there is no colour change after 120 seconds, record as 'more than 120'.

   (iv) Record your results in an appropriate table.

[4]

You are required to estimate the concentration of reducing sugars in **C1**, **C2** and **C3**.

4. Carry out the Benedict's test on each of **C1**, **C2** and **C3** and record the time taken for the appearance of the first colour change.

   (v) State the colour appearance and time taken for appearance of first colour change for **C1**, **C2** and **C3.**

   Colour appearance

   C1 .... **C2** .... **C3** .... [1]

   Time taken

   C1 .... **C2** .... **C3** .... [1]

5. The method described from steps 1 to 4 might not give an accurate measurement of the sugar content in the unknown samples **C1** to **C3**.

(v) Explain why.

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…………………………………………………………………………………………..…[1]

(vi) Describe the test that should be done to give an accurate measurement on the presence of sugars in the unknown samples.

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(b) Some students studied the stickiness of cotton fibres using the colorimetric method. A colorimeter can be used to measure the quantity of light absorbed (absorbance) by a solution. The data can be used to calculate cotton stickiness index using the colour index obtained from the samples after subjected to various processes. A high quality cotton fibre has a low stickiness index.

A calibration graph was drawn by plotting the stickiness index against the known concentrations of sugar content in the six solutions. The graph can be used to estimate the concentration of sugar in an unknown sample, **U**. Table 2.2 shows the results for the stickiness index of solutions of known sugar concentration.

Table 2.2

Sugar content / μg cm^{-3}	**Stickiness index**
0.2	1.2
0.4	2.2
0.6	3.0
0.8	3.9
1.0	4.9

(i) Use the grid to display the results shown in Table 2.2 in an appropriate form.

[4]

(ii) The student found that the stickiness index of the unknown sample **U,** is 4.2. State the sugar content and suggest why the cotton output is low.

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(c) Gossypol is a natural phenol derived from the cotton plant. Gossypol is a phenolic aldehyde that permeates cells and acts as an inhibitor for several dehydrogenase enzymes.

Lactate dehydrogenase catalyses the conversion of pyruvate to lactate. NADH is re-oxidised back to NAD^+. Gossypol can inhibit lactate dehydrogenase.

The extent of inhibition depends on the concentration of gossypol. A student carried out an investigation and found that lactate dehydrogenase was completely inhibited by gossypol at concentrations of gossypol greater than 5.0%.

The student hypothesised that concentration of gossypol below 5.0% will continue to inhibit the enzyme.

Design an experiment to investigate the effect of gossypol concentration on the rate of lactate dehydrogenase activity, using DCPIP as an indicator.

Reduced DCPIP is colourless and when oxidised, forms a blue compound.

In your plan, you must use:

- 5.0% gossypol solution
- 1.0% lactate dehydrogenase solution
- 1.0% pyruvate solution
- 1.0% DCPIP solution.
- Distilled water
- A thermostatically-controlled water bath set at 35°C
- A pH 6.5 buffer solution
- Colourimeter and cuvette

You may select from the following apparatus and plan to use appropriate additional apparatus:

- normal laboratory glassware, e.g. test-tubes, boiling tubes, beakers, measuring cylinders, graduated pipettes and pipette fillers, glass rods, etc.
- syringes
- timer, e.g. stopwatch

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- be illustrated by relevant diagram(s), if necessary
- identify the independent and dependent variables
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible
- include a layout of results tables and graphs with clear headings and labels
- use the correct technical and scientific terms
- include any reference to safety measures to minimise any risks associated with the proposed experiment

[9]

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[Total: 30]

3 Fig. 3.1 is a photomicrograph of a stained transverse section through a cotton leaf.

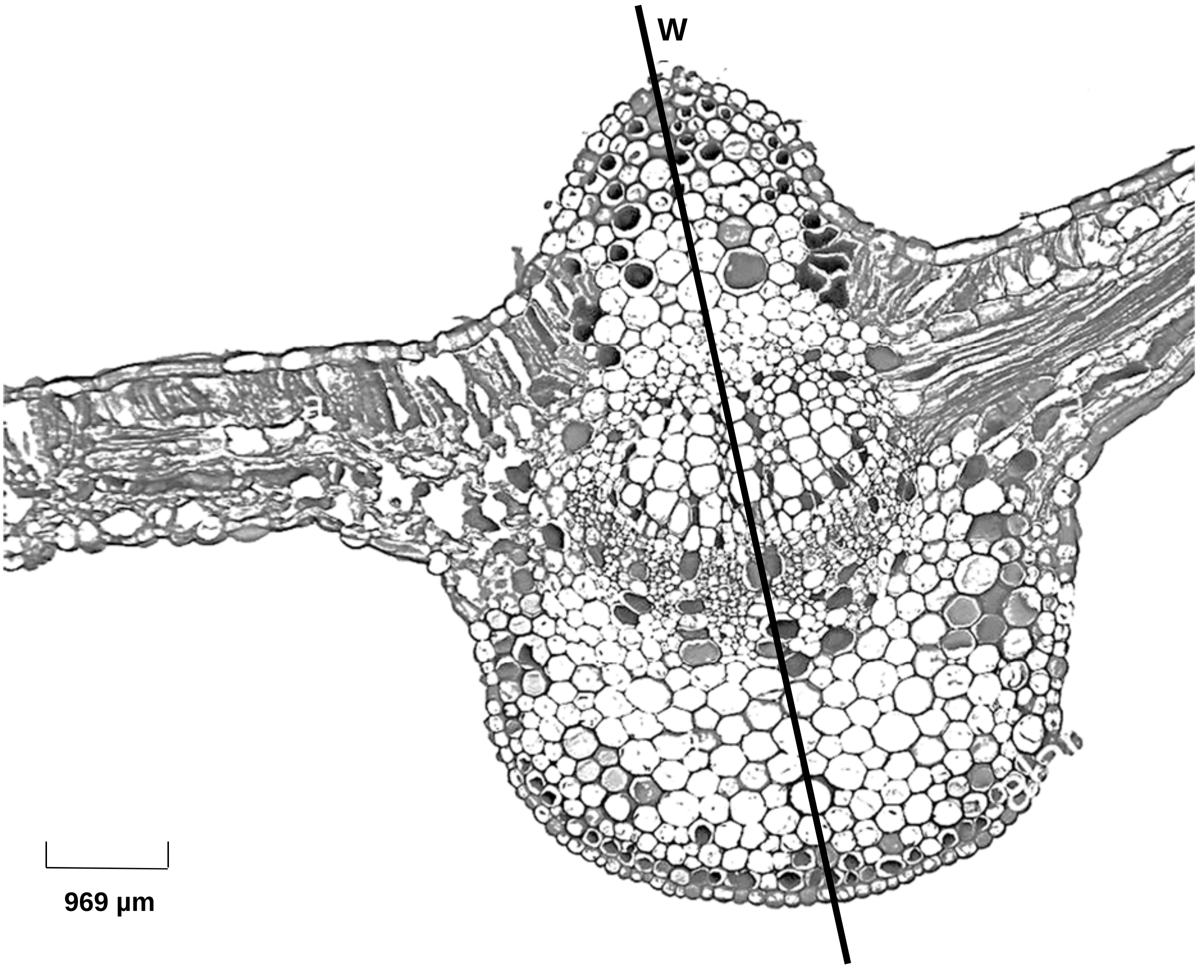

Fig. 3.1

(a) You are required to use a sharp pencil for drawings.

Draw a large plan diagram of the section of the leaf in Fig. 3.1, as shown by the shaded area in Fig. 3.2, in the space provided on page 19. A plan diagram only shows the arrangement of the different types of tissues. Individual cells must **not** be drawn in plan diagrams.

Use **one** ruled label line and label your diagram to identify the vascular bundle.

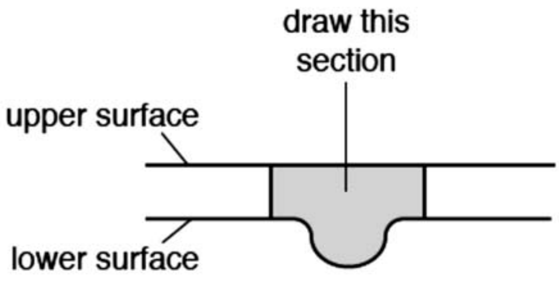

Fig. 3.2

[5]

(b) (i) **P1** is a slide of a stained transverse section of cells taken from the lymph gland.

Observe the cells in **P1** under x400 magnification and the xylem cells in Fig. 3.1 and identify two differences between them.

[2]

(ii) Calculate the **actual** width of the leaf at the position marked by line **W**, using the scale bar provided. You may lose marks if you do not show your working or if you do not use appropriate units.

Actual width of leaf .... [3]

[Total: 10]